Case Study

From G-code to Part to Prediction: DED Simulation with Ansys Discovery and Ansys Mechanical software

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Ansys Software Used

This resource uses Ansys Discovery™ 3D product simulation software and Ansys Mechanical™ structural finite element analysis software

Summary

Directed Energy Deposition (DED) additive manufacturing (AM) offers significant design flexibility for complex components, but strong thermal gradients during the process can lead to distortion and residual stresses. This case study presents a process-level simulation workflow for DED within the Ansys ecosystem, combining path-based geometry reconstruction with weakly coupled thermal–mechanical analysis.

A multi-axis G-code describing the deposition path is first used in the Additive DED add-in of Ansys DiscoveryTM to generate a path-accurate geometric representation of the part. This model is subsequently used in Ansys MechanicalTM to perform a weakly coupled simulation of the build process. The approach captures the evolution of temperature, distortion, and residual stresses throughout deposition and cooldown.

The modelling strategy focuses on macro-scale thermal and mechanical behaviour, enabling efficient prediction of global manufacturability trends while intentionally neglecting fine-scale phenomena such as melt pool dynamics and microstructural evolution. Despite this simplification, the results provide valuable insight into the sensitivity of distortion to process parameters and build strategies.

Overall, the study demonstrates how simulation-driven workflows can support design for additive manufacturing (DfAM), enabling early identification of potential print failures and facilitating informed decisions to improve part quality and process robustness.

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1. Introduction

Directed Energy Deposition (DED) is an additive manufacturing (AM) process in which focused thermal energy – typically from a laser, electron beam, or plasma arc – is used to melt feedstock material as it is being deposited (see Figure 1). DED is also known as laser engineered net shaping; electron beam additive manufacturing, wire arc additive manufacturing, or laser deposition technology. It encompasses the process in which a laser or electron beam creates a melt pool on previously solidified material where blown powder or fed wire is used. Like a powder bed fusion (PBF) process, the DED processes are characterized by high temperatures applied locally and consequently severe thermal gradients. These lead to overheating, distortions and eventually to the buildup of residual stresses as the layers are deposited. However, unlike PBF, DED systems often operate with multi-axis capabilities, enabling material to be added in complex orientations or onto existing components (see Figure 2). This makes DED particularly valuable for repair, feature addition, and the fabrication of large, functionally optimized structures. As it commonly produces a rougher surface compared to PBF, the as-printed parts are often generated with a generous padding (near-net-shape), so it can subsequently be machined to achieve a smooth surface and nominal dimensions.

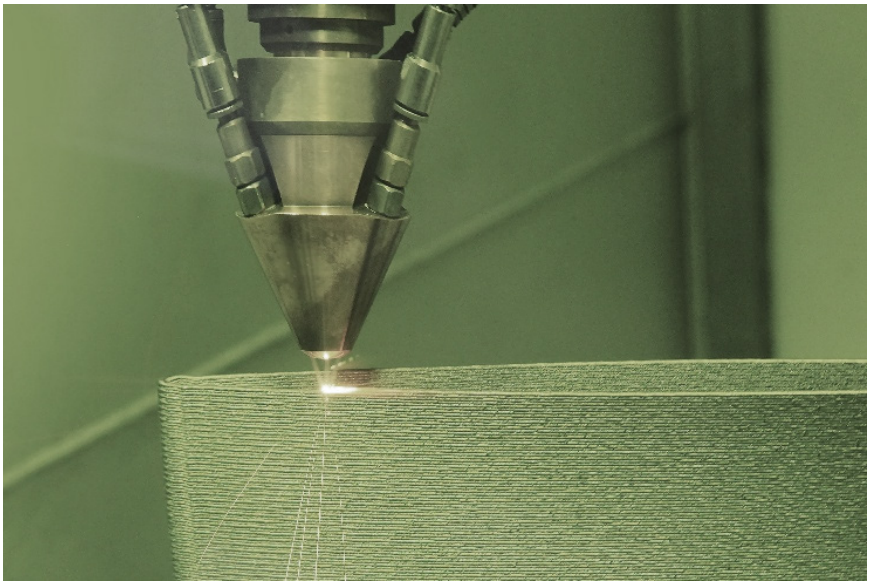


Figure 1: Directed Energy Deposition process showing the print head applying material layer-by-layer.

In this case study, the additive manufacturing of a logo printed onto a substrate (e.g. part or print bed) using DED will be simulated. First, the toolpath is used to recreate the 3D model, which is then linked with the processing parameters to simulate the process to identify any potential print failures before printing.

1.1 Path Planning in DED

At the heart of any AM process is the toolpath, most commonly defined through G-code. In multi-axis systems, this code controls not only the linear motion of the deposition head but also the rotation and tilt of additional axes. The G-code therefore represents the manufacturing “plan”: the precise sequence and orientation of material placement, deposition rate, and energy delivery. Understanding this plan is crucial, because the path directly determines the heat input distribution, the timing of material deposition, and ultimately the microstructure and mechanical response of the part. The order

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in which layers and beads are deposited governs:

1. Local and global heat accumulation
2. Thermal gradients and cooling rates
3. Residual stresses
4. Distortion and warpage
5. Potential defect formation

As such, accurately representing the real deposition path in simulation is essential for meaningful predictions.

1.2 DED Process Simulation

DED involves rapid, localized heating followed by relatively slow cooling as subsequent material is added. This creates non-uniform expansion and contraction throughout the part. A coupled transient thermal–static structural simulation is required because:

1. The temperature field changes over time as material is deposited and heat is applied,
2. Thermal gradients drive thermal strains, which produce stresses and distortion,
3. The structural response depends strongly on the evolving thermal load, and
4. Predicting final shape, residual stress, and potential failure locations relies on resolving these time-dependent interactions.

The objective of the coupled transient-thermal and structural simulation in Ansys Mechanical software is to evaluate the global thermal and mechanical response of a print model manufactured using a DED process. The analysis focuses on predicting temperature-induced distortions and residual stresses that develop during the build and cooling phases, which can compromise geometric accuracy, balance, and structural integrity of the printed model during and after the build. Thus, a simulation of the DED process helps with making targeted changes to the CAD-model and/or process parameters, etc. to achieve a first-time-right-print (see Figure 2).

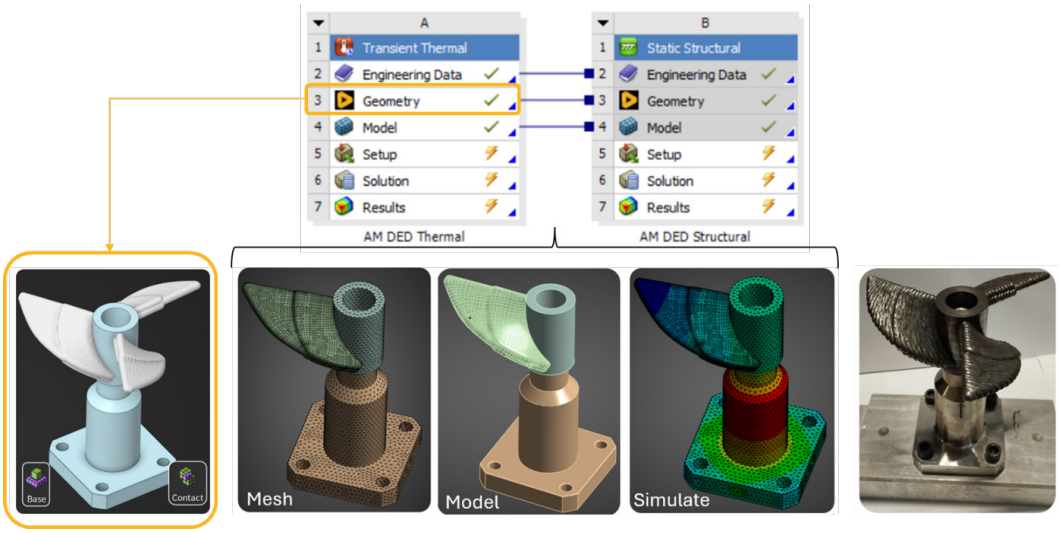


Figure 2: Workflow from path to printed part, utilizing part preparation and process simulation capabilities, exemplified by a propeller blade printed onto a substrate [1].

Rather than resolving fine-scale phenomena such as melt pool behavior or microstructural evolution, the simulation captures macro-level trends that influence manufacturability. These results are used to

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assess how process choices—including build orientation, deposition sequence, and support from the base plate—affect the final shape and stress state of the printed model.

The simulation is intended as a design-for-additive-manufacturing (DfAM) tool, providing insight into how changes in geometry or process setup may reduce distortion and improve print success. Detailed modeling of melt pool physics or microstructure development is outside the scope of this analysis and would require specialized high-fidelity thermal-fluid simulations.

1.2.1 Thermal Analysis

In a DED simulation, the thermal behavior of the process is governed primarily by the presence of a moving heat source that represents the laser or energy beam. This heat source must be defined with parameters such as its power, efficiency, and beam radius so that the simulation can accurately reproduce localized heating as the deposition head travels along the toolpath. Once energy is introduced into the material, transient heat conduction distributes this heat through both the freshly deposited layers and the underlying substrate. Because all exposed surfaces exchange heat with the environment, convection and radiation boundary conditions must be applied to allow realistic cooling behavior.

As the deposition progresses, new material appears over time, the simulation must alive (heat-up) elements gradually, reflecting the physical sequence of bead and layer creation encoded in the G-code. Heat input during the deposition process can be represented using two alternative approaches: an applied temperature method or a power-based heat generation method. In the applied temperature approach, each newly activated cluster (a set of mesh elements splitting the weld lines into smaller volumes) is assigned a prescribed temperature at deposition (please refer to Section 2.2.2 or the documentation [2] for more details). This formulation assumes that process parameters are selected such that the material temperature remains at or above the melting point, ensuring proper fusion, while not significantly exceeding it, thereby avoiding excessive overheating or keyholing effects. Alternatively, in the power-based approach, heat is introduced through a volumetric heat generation load applied to newly activated elements. This load is calculated from the input laser power, material absorptivity, and deposition rate, and distributed over a defined simulation time interval. The resulting volumetric heat generation (HGEN) therefore reflects the total energy delivered to each cluster during deposition, with the effective application time controlled indirectly through the thermal calibration settings:

$$\mathrm{HGEN} = \frac{\mathrm{P}\eta}{\mathrm{d}\mathrm{t}_{\mathrm{a}}} \tag{1}$$

where:

- HGEN=heat generation load
- P= laser power
- η=Absorptivity
- d=deposition rate
- t_a=simulation loading applied time

To accurately describe how the material responds as it is heated and subsequently cools, temperature-dependent thermal properties (such as thermal conductivity and specific heat) must be included.

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With the AlSi10Mg used in this study, a Ansys predefined AM-material [3] was used that includes – among other things – the bilinear isotropic hardening [4] plasticity of the material. Alternatively, these properties could be obtained from the JAHM temperature-dependent material data [5], which is available in Ansys Granta Selector or Ansys Granta MI tools, or from calibrating the properties from experiments.

1.2.2 Structural Analysis

The mechanical response of the structure arises directly from the evolving temperature field. As the material is heated, it undergoes thermal expansion, and as it cools, it contracts; these thermal strains drive the formation of stresses and ultimately determine the deformation state of the final part. To capture these effects properly, the simulation uses temperature-dependent mechanical properties, including elastic–plastic behavior that evolves with the thermal history (i.e. material properties must capture the phase transformation of the printed material). For some high-temperature alloys, it may also be necessary to incorporate creep or stress relaxation, ensuring that stress reflects both instantaneous and time-dependent deformation mechanisms.

Boundary conditions play a critical role in determining how stress accumulates. If the component is clamped to a build plate, fixed or partially constrained supports must be defined to represent those fixturing conditions. If symmetry can be exploited, appropriate symmetry constraints may be applied; otherwise, unconstrained surfaces must remain free to deform, enabling prediction of the warpage commonly observed in DED parts. Contact or interaction definitions may also be required wherever the part interfaces with the substrate or other features in the model.

1.2.3 Coupled Analysis

Because thermal and mechanical phenomena in DED are tightly interconnected, the simulation typically proceeds through a sequentially (weakly) coupled thermal–mechanical workflow. In this approach, the thermal analysis runs first, computing the transient temperature field associated with each step of the deposition process. The simulation must use time increments that correspond to the actual deposition rate and the movement of the heat source along the toolpath, ensuring that the heating and cooling patterns match the physical process.

The resulting temperature distribution is then passed as a load into a structural analysis, which calculates stresses, strains, and displacements for the same time intervals. To maintain consistency between the two analyses, the element activation schedule in the structural model must mirror the activation schedule used in the thermal model, so that material appears mechanically only when it has been thermally introduced. This alignment ensures that the thermal history of each newly added bead directly influences the resulting deformation and residual stress state.

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2. Method

The workflow from DED toolpath to process simulation start in Ansys Workbench and utilizes the inbuild Custom System called “AM DED Process” which automatically couples a transient thermal with a static structural analysis system (see Figure 2), i.e. uses Ansys Mechanical software as solver, which covers the dedicated metal additive manufacturing process simulation of Ansys Additive Suite tools.

As part of the pre-processing stage, the engineering data as well as the CAD model must be defined (see cell A2 and A3 in Figure 2). For the former the inbuild AM material database was used, and it is assumed that the print model will be manufactured from AlSi10Mg, which is a commonly used aluminum alloy in metal-AM. It should be noted that the process parameters defined for each material later in Section 2.2, is greatly affecting the part’s performance.

Subject of investigation is a cylinder onto which a spiral print track is printed (see Figure 3). The geometry preparation is conducted in Ansys Discovery software. Its Additive DED add-in [6] enables path-to-part modeling, whereby the import of G-code or similar machine instructions are used to reconstruct the true deposition geometry—bead by bead, layer by layer. This geometry can then be transferred into Ansys Mechanical software for a coupled thermal–mechanical analysis that captures the evolving physics throughout the build.

2.1 G-code to Part

In this study, the deposition process of a DED build represents a spiral print track printed onto a cylindrical substrate or base (e.g. fins on a heat sink). It is defined using a standard ASCII formatted G code file, which serves as the foundation for geometry reconstruction and simulation. G code describes the toolpath as a sequence of explicit machine instructions; thus, encoding the order, timing, and orientation of each deposited bead, forming an accurate digital representation of the manufacturing process that can be directly interpreted by the Additive DED add-in [6] (see Figure 3). For this use case a circular bead geometry with 2.5 mm bead radius was chosen. Lastly, the contact between the print track and the base was identified and converted to a named selection to ease the contact definition in the subsequent simulation setup (see section 2.2).

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Figure 3: Print model geometry preparation from G-code to part and subsequently to a simulation-ready setup using Ansys Discovery's Additive DED add-in.

The multi-axis G-code snippet in Figure 4, represents the start of the spiral print manufacturing program. Material deposition itself is defined by using linear interpolation moves with specified feed rates (neglected here). Therefore, each command line includes X, Y, and Z, coordinates, defining the physical trajectory of the deposition head while maintaining constant process conditions. Of particular importance for simulation is the distinction between deposition active and travel segments, i.e. the laser on/off indicators. These laser on/off indicators are the primary triggers used by the Discovery Additive DED add in to determine when material is being added. In this printed model example, deposition begins with G1 (laser on) and ends with G0 (laser off). In this example, a single layer is printed continuously, however, this clear separation between active and inactive segments is even more crucial for defining element activation schedules and for driving a moving heat source in the subsequent mechanical analysis. There are a number of supported commands [7] in Ansys Mechanical software which form the basis for the subsequent DED process simulation. It should be noted that exact syntax and availability of codes vary by controller and DED vendor. In many retrofitted systems, the "spindle" is repurposed to control laser power; in others, a dedicated M code (manages auxiliary machine functions, e.g., spindle on/off, tool changes, coolant) or analog channel drives power.

SpiralCylinderGcode

```
1   G0 X50.0 Y0.0 Z0.0
2   G1 X49.11082077026367 Y9.387618064880371 Z0.20040079951286316
3   G1 X46.47490692138672 Y18.44134521484375 Z0.4008015990257263
4   G1 X42.18600845336914 Y26.83916473388672 Z0.6012024283409119
5   G1 X36.396671295166016 Y34.28239059448242 Z0.8016031980514526
6   G1 X29.312808990478516 Y40.50628662109375 Z1.0020040273666382
7   G1 X21.186370849609375 Y45.289485931396484 Z1.2024048566818237
8   G1 X12.30639362335205 Y48.46186828613281 Z1.4028055667877197
9   G1 X2.988711357116699 Y49.91059494018555 Z1.6032063961029053
10  G1 X-6.4352707862854 Y49.584144592285156 Z1.8036072254180908
11  G1 X-15.63036823272705 Y47.49412155151367 Z2.0040080547332764
12  G1 X-24.2695369720459 Y43.71
```

Figure 4. Multi-axis G-code (machine control commands) snippet for the print model.

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2.2 DED Simulation Setup

The DED Process Wizard [8] was used to set up the simulation, simplifying the clustering, mesh generation, DED contact definition, automatic named selection creation, machine parameter setting and boundary condition definition.

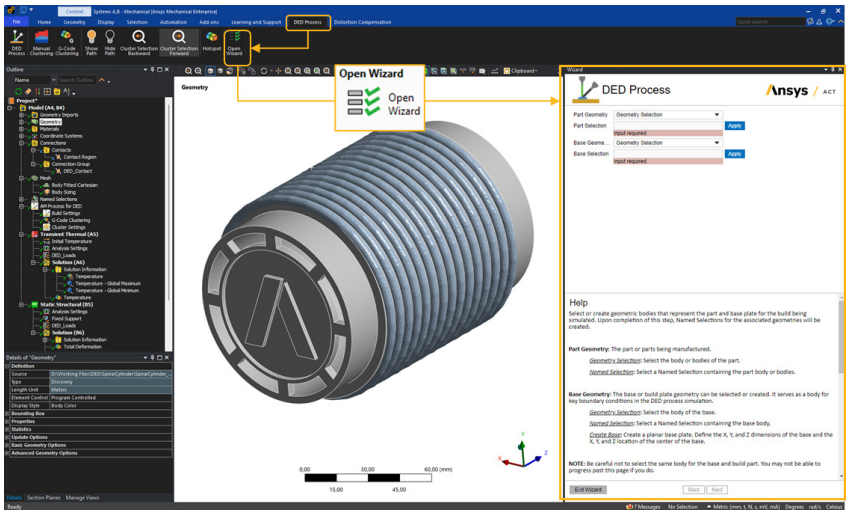


Figure 5. DED Process Wizard in Ansys MechanicalTM, simplifying the process simulation setup.

2.2.1 Meshing

For the DED simulation the mesh should capture the weld track correctly, but it does not necessarily require a layered mesh with identical layer height for every layer. Here a Cartesian mesh with Hex8 (linear) elements was chosen for the spiral weld track with an element size of 0.85 mm, which ensures a balance between computational time while avoiding a too course representation of the spiral which would cause the gaps of each turn to coalesce (see Figure 6). Concurrently, this ensures a good element quality and aspect ratio; the latter is critical for numerical convergence during the static structural simulation. The cylindrical substrate (base) was meshed with Tet4 elements with a mesh size of 4 mm. As a result of the meshing using the DED-Wizard [8], all relevant named selection for contact and clustering are automatically generated.

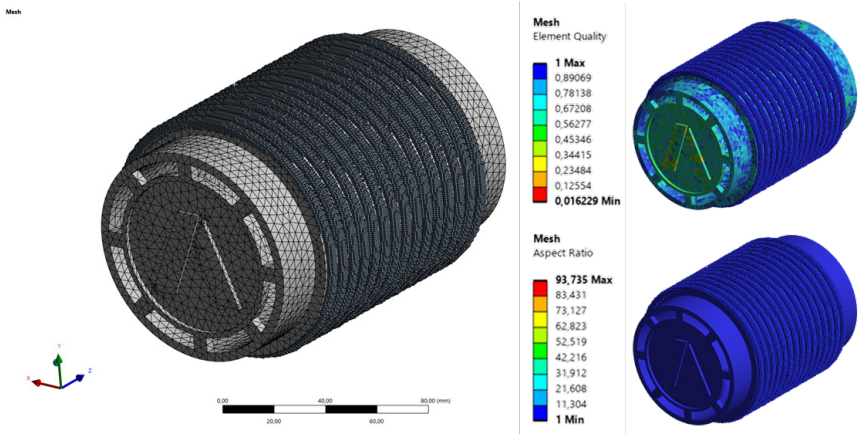


Figure 6. Meshed DED-model with tetrahedral elements for the cylindrical substrate and cartesian hexahedral elements for the spiral, showcasing good overall element quality and aspect ratio.

In the DED process, a special case of bonded contact between the element faces on the bottom of the build and the element faces on the top of the base is used, whereby the behavior is set to ‘Asymmetric’

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with ‘Nodal-Normal from Contact’ as the detection method.

2.2.2 Build Settings and G-Code Clustering

The DED process simulation follows the actual build sequence of the part, activating material progressively along the deposition path, using an element birth and death approach [9]. For the DED simulation this means that all elements are initially deactivated and become sequentially reactivated, i.e. material is added in discrete portions called element clusters, each representing a short segment of the deposition path. In a deactivated state, the elements properties are multiplied by a reduction factor of 1.0E-6 and their loads, strains or specific heats are zeroed out. Once elements are ‘born’, i.e. a part of the weld track is created, the reactivated elements’ (i.e. are part of a cluster volume) stiffness, mass, loads, etc. return to their full original values while being strain-free (no history record). When elements are newly activated, their thermal strains are determined from the difference between the current load step temperature and the reference temperature. This can lead to unintended stresses, meaning that such elements may not be stress-free despite being newly introduced.

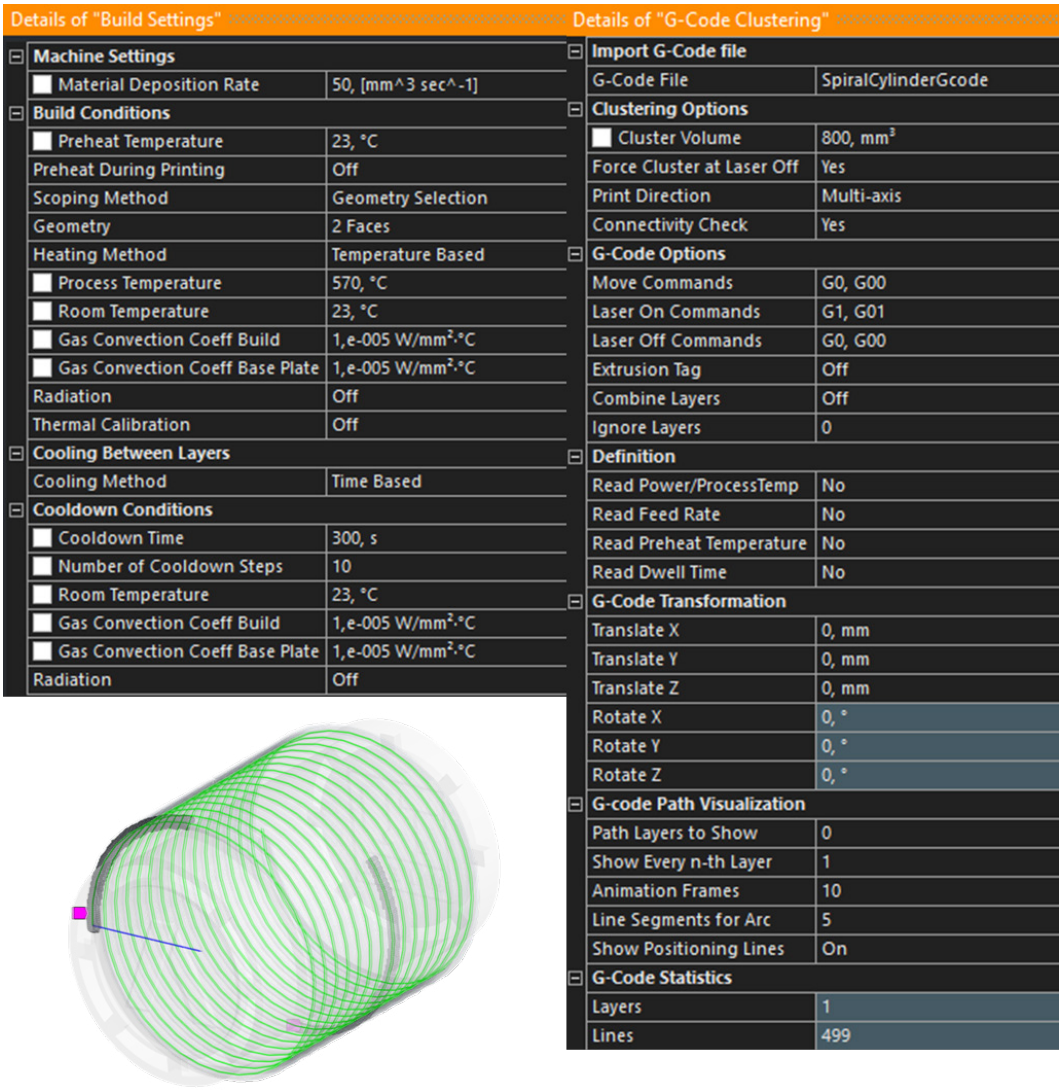
Thermal gradients and cooling behavior are modeled at a macro scale, providing sufficient accuracy to predict overall distortion trends. The time integration scheme employs relatively large time steps, which adequately capture thermally induced strains but do not resolve detailed heating and cooling cycles at smaller scales.

Here a temperature-based approach was employed, meaning each newly activated cluster is assigned a prescribed process temperature at the moment of deposition. This approach assumes that the process parameters are chosen such that:

1. The deposited material reaches at least the melting temperature, ensuring proper fusion between layers
2. The temperature does not significantly exceed the melting point, thereby avoiding excessive overheating or phenomena such as keyholing

For this DED simulation a cluster volume of 800 mm^3 was used (equal to approximately a quarter revolution of the spiral) with a deposition speed of 50 mm^3/sec, resulting in 45 clusters for this single layer spiral track. For clustering, the multi-axis G-code (recall Section 2.1) was read-in with the print and move commands G1 and G0 defined as laser on/off actions. It should be noted that these cluster volume and process parameters are key options available to tune the simulation. Figure 7 summarizes the process and cluster settings used with the environmental parameters characterizing the thermal conditions surrounding the part throughout the deposition process. Distinct convection coefficient and emissivity values can be assigned, allowing for differentiation between the material properties of the deposited part and the base plate. Here, default settings were used, and a cooldown time of 300 sec was employed after all clusters were activated. For further details on individual parameters, the reader is kindly referred to the documentation, specifically [10].

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Details of "Build Settings"	
Machine Settings	
Material Deposition Rate	50, [mm^3 sec^-1]
Build Conditions	
Preheat Temperature	23, °C
Preheat During Printing	Off
Scoping Method	Geometry Selection
Geometry	2 Faces
Heating Method	Temperature Based
Process Temperature	570, °C
Room Temperature	23, °C
Gas Convection Coeff Build	1,e-005 W/mm²·°C
Gas Convection Coeff Base Plate	1,e-005 W/mm²·°C
Radiation	Off
Thermal Calibration	Off
Cooling Between Layers	
Cooling Method	Time Based
Cooldown Conditions	
Cooldown Time	300, s
Number of Cooldown Steps	10
Room Temperature	23, °C
Gas Convection Coeff Build	1,e-005 W/mm²·°C
Gas Convection Coeff Base Plate	1,e-005 W/mm²·°C
Radiation	Off

Details of "G-Code Clustering"	
Import G-Code file	
G-Code File	SpiralCylinderGcode
Clustering Options	
Cluster Volume	800, mm³
Force Cluster at Laser Off	Yes
Print Direction	Multi-axis
Connectivity Check	Yes
G-Code Options	
Move Commands	G0, G00
Laser On Commands	G1, G01
Laser Off Commands	G0, G00
Extrusion Tag	Off
Combine Layers	Off
Ignore Layers	0
Definition	
Read Power/ProcessTemp	No
Read Feed Rate	No
Read Preheat Temperature	No
Read Dwell Time	No
G-Code Transformation	
Translate X	0, mm
Translate Y	0, mm
Translate Z	0, mm
Rotate X	0, °
Rotate Y	0, °
Rotate Z	0, °
G-code Path Visualization	
Path Layers to Show	0
Show Every n-th Layer	1
Animation Frames	10
Line Segments for Arc	5
Show Positioning Lines	On
G-Code Statistics	
Layers	1
Lines	499

Figure 7. Build and cluster settings used for DED print simulation and visualization of print path (green) and two cluster volumes along the path.

3. Results and Discussion

3.1 Transient Thermal Solution

As shown in Figure 8 by the graphs of the transient thermal analysis, each cluster is heated up to the process temperature of 580°C followed by 10 sub steps defining its cool down (compare Figure 7). After each cluster was reactivated the final cooldown is represented by the drop of the temperature curve at the end.

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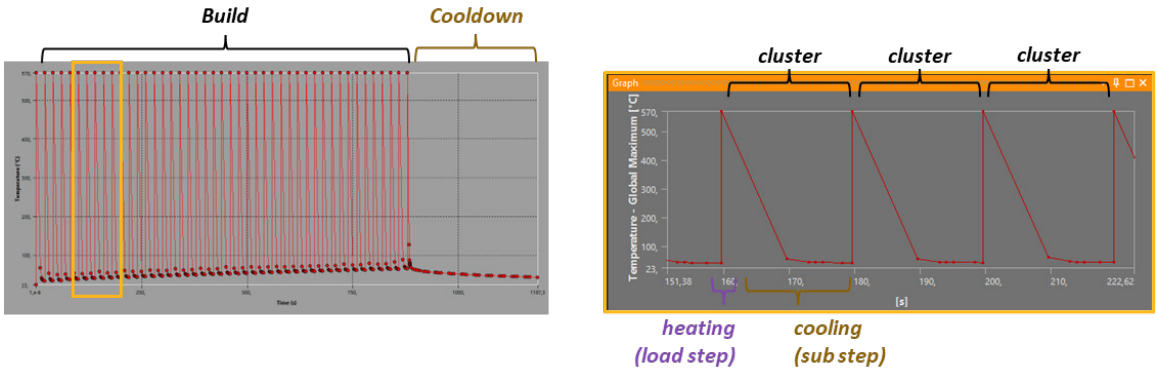


Figure 8. Temperature graphs of the transient thermal analysis, highlighting the 43 clusters being sequentially activated in each load step of the build followed by a cooling sub step before a final cooldown phase.

The resultant temperature distribution at the end of the cooldown and during the print process is depicted in Figure 9. The image shows how the spiral is printed in positive z-direction, causing a final heat distribution that increases with +z. After the cooldown phase of 300 sec, the part has not yet reached the ambient temperature but is ready for extraction by hand from the machine. It should be noted that a cluster volume is chosen that approximately represents a quarter of a turn and is activated at once. For a more accurate result of the print process, a smaller cluster volume could be chosen at the expense of longer runtimes, since the time for a load step is determined by this volume divided by deposition rate.

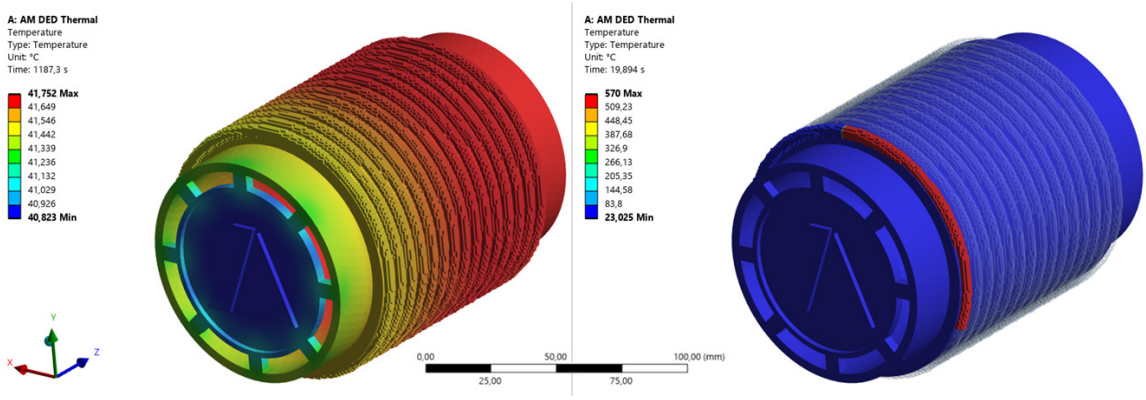


Figure 9. Temperature distributions at the end of the cooling stage (left) and during the print and activation of a cluster (right).

3.2 Static Structural Solution and Distortion Compensation

Figure 10 showcases the final deformations and strains in the spiral and final part, respectively. With this representing a single track without any changes in the toolpath (strategy) or changes of the process parameters during the process, the deformations and strains are homogenous across the part with a maximum deformation of 0.25 mm and less than 1% elastic strain. The average deformation in the part is in the range of 4e-2 mm.

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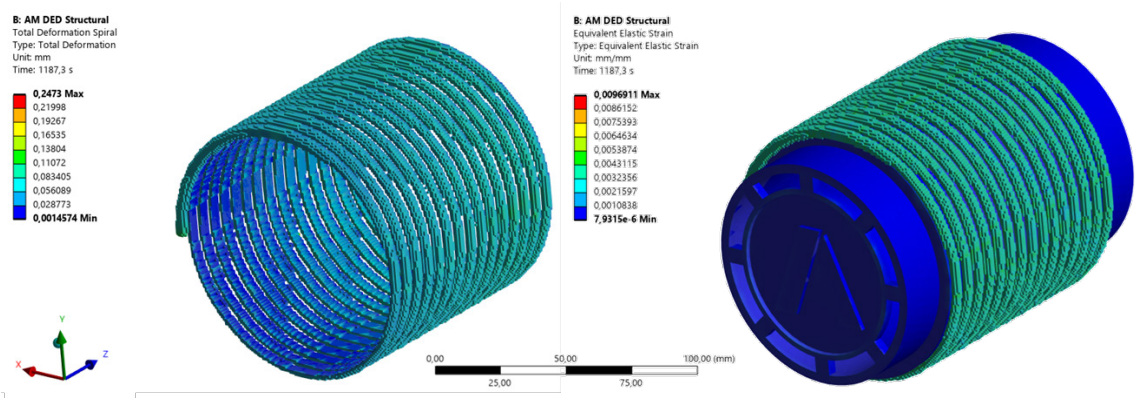


Figure 10. Deformation of the spiral (left) and elastic strain of the final part (right) after the cooldown phase.

While these process-induced distortions are generally acceptable, there are instances where these may cause out-of-tolerance parts that should be avoided. This can for instance be the case if higher deposition speeds are used to increase throughput or if thermally dissimilar print and base materials are used. A single-run distortion compensation [11] was conducted for the spiral print which yields a pre-distorted (compensated) geometry through reverse-distorting the single static structural result (see Figure 11). Given the mesh size used, a minimum curvature size of 0.5 mm and a maximum size of 1 mm were used for creating the compensated STL. After importing the compensated STL and the original STL of the spiral into Ansys Discovery program, the deviation tool was employed, revealing maximum compensations at the start/end of the track of about 0.07 mm and some minor positive and negative deformation on the outside and contact faces of the spiral, respectively.

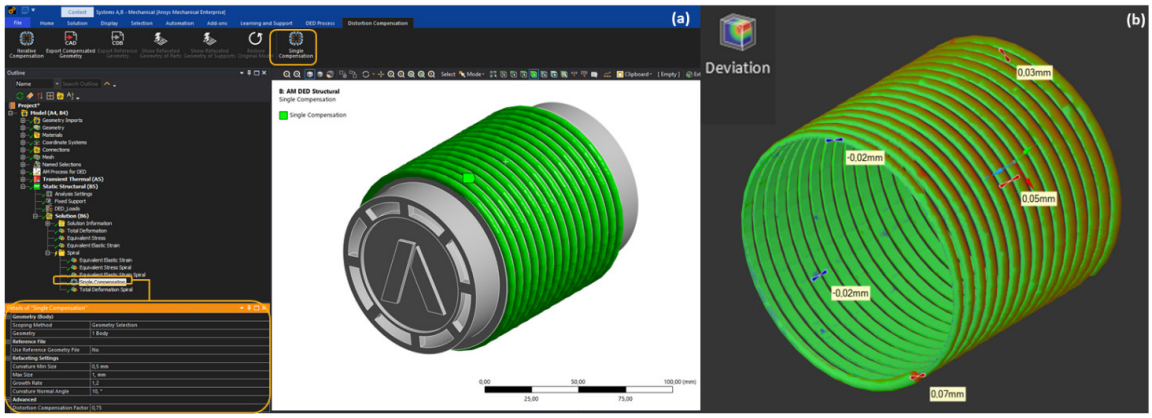


Figure 11. (a) Ansys Additive Solutions' distortion compensation add-in Ansys Mechanical software. (b) Magnitude of deviation between the original and compensated STL, assessed with the deviation tool in Ansys Discovery.

4. Conclusion

This case study demonstrates a practical workflow for simulating Directed Energy Deposition (DED) processes using a path-based approach that links G-code, geometry reconstruction, and coupled thermal–mechanical analysis. By focusing on macro-scale temperature evolution and structural response, the simulations provide valuable insight into distortion trends and residual stress development during multi-axis deposition processes.

The results highlight that, even with simplified thermal modeling and coarse time discretization, the approach is sufficient to capture global deformation behavior and identify potential manufacturability

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risks. Furthermore, the integration of clustering strategies and temperature-based material activation enables an efficient representation of the build process while maintaining computational tractability. Finally, the use of distortion compensation illustrates how simulation-driven design adjustments can improve dimensional accuracy and reduce the likelihood of build failure. Overall, the presented methodology supports design-for-additive-manufacturing (DfAM) by enabling informed decisions on process parameters, build strategies, and geometry modifications, ultimately contributing to more reliable and first-time-right DED production.

5. References

[1] Ansys, Inc.: Simulate With Real Geometry: Path-Based Modeling for Additive Applications. 2026.
[2] Ansys, Inc.: Directed Energy Deposition Simulation Guide - 5.8. Perform Element Clustering
[3] Ansys, Inc.: Additive Manufacturing 2026 R1 - 5.2. Define Engineering Data. 2026R1
[4] Ansys, Inc.: Ansys Product Help - 4.4. Rate-Independent Plasticity. 2026R1.
[5] Ansys, Inc.: Additive Manufacturing 2026 R1 - 6.5. Using JAHM Temperature-Dependent Material Data in AM Simulation. 2026R1
[6] Ansys, Inc.: Directed Energy Deposition Simulation Guide - 3.2. Creating a Part from a G-code File: Path to Part. 2026R1
[7] Ansys, Inc.: Directed Energy Deposition Simulation Guide - Chapter 6: Supported G-Code Commands. 2026R1
[8] Ansys, Inc.: Directed Energy Deposition Simulation Guide - Chapter 4: Using the DED Process Wizard
[9] Ansys, Inc.: Directed Energy Deposition Simulation Guide - 8.2. Understanding Element Birth and Death. 2026R1.
[10] Ansys, Inc.: Directed Energy Deposition Simulation Guide - 5.7. Define Build Settings. 2026R1
[11] Ansys, Inc.: Distortion Compensation Guide - 2.1. How Single Compensation Works. 2026R1

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